
THE SHOT CLOCK: TIMING A CLEVELAND SELLOUT

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TABLE OF CONTENTS

01

PROBLEM OVERVIEW

Steps we will take to help the
Cavs maximize ticket sales

02

PACING LINE & DANGER ZONE

Visually representing how the
Cavs can get to their 2500 ticket
goal per game

03

MODELING TICKET SALES

Predicting ticket sales from 20
days out for a single game

04

CONCLUSIONS & RECOMMENDATIONS

Explaining a dashboard of
results and giving limitations
and recommendations of model

01

PROBLEM OVERVIEW

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GOAL: Help the Cavs maximize ticket sales for a single game

- How are we going to do this?
 - Develop a ticket Pacing Line
 - Create a Pacing Line based off historical sales in each ticket tier
 - Add Danger Zones
 - Create two Danger Zones → one for when to market and another for when to reduce prices
 - Predict sales for the Cavs vs. Hawks on November 2, 2025
 - Use 5-fold cross validation and predict ticket sales from 20 days out
 - Create a dashboard summarizing results
 - Utilize R Shiny to create an easy-to-interpret dashboard



DATA OVERVIEW

- Original data consists of individual ticket sale transactions for the 2023-24 and 2024-25 Cavs seasons
 - The game we aim to predict is for this year's season
- Pre-processing steps:
 - Split tiers into 4 categories, due to low sample size in both Tiers A+ and E
 - Tier1 → Tier A+ and Tier A (15 games)
 - Tier2 → Tier B (37 games)
 - Tier3 → Tier C (37 games)
 - Tier4 → Tier D and Tier E (33 games)
 - Created binary values to simplify categorical variables (weekend, giveaway, etc.)
 - From 60 days out, calculated cumulative ticket sums for each day leading up to the game

EXAMPLE DATA

GAME	DAYS UNTIL GAME	CUMULATIVE TICKETS SOLD	TICKETS SOLD ON DAY	GIVEAWAY?	SPECIAL JERSEY?	THEME NIGHT?	WEEKEND?	DAY_OF_WEEK	TIER GROUP	TIER
CAV23_22_1023 - Cavaliers vs. Washington Wizards	60	320	97	1	0	1	1	Sunday	Tier2	Tier B
CAV23_22_1023 - Cavaliers vs. Washington Wizards	59	329	9	1	0	1	1	Sunday	Tier2	Tier B
CAV23_22_1023 - Cavaliers vs. Washington Wizards	58	342	13	1	0	1	1	Sunday	Tier2	Tier B
CAV23_22_1023 - Cavaliers vs. Washington Wizards	57	360	0	1	0	1	1	Sunday	Tier2	Tier B
CAV23_22_1023 - Cavaliers vs. Washington Wizards	56	360	18	1	0	1	1	Sunday	Tier2	Tier B

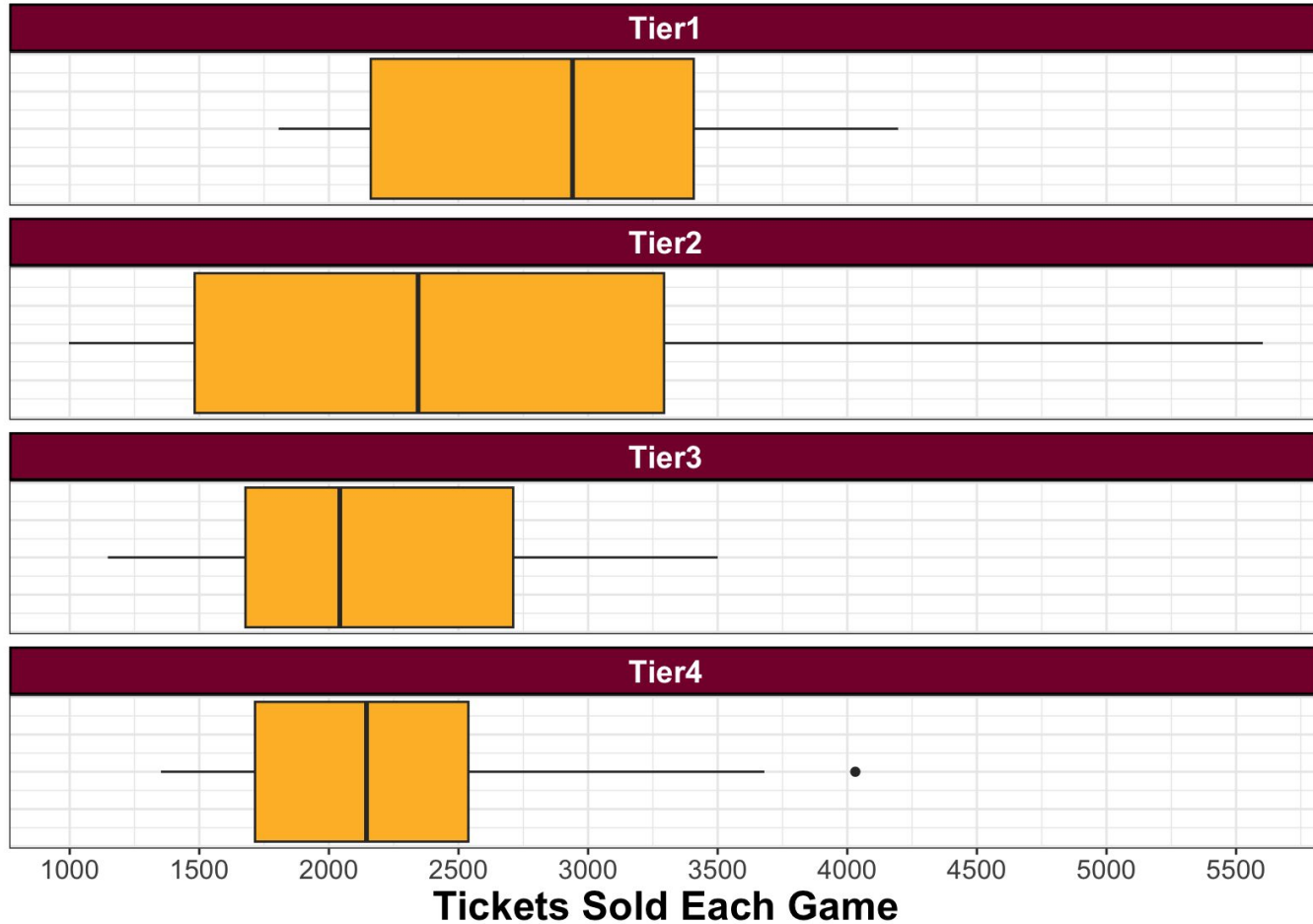
02

PACING LINE & DANGER ZONE

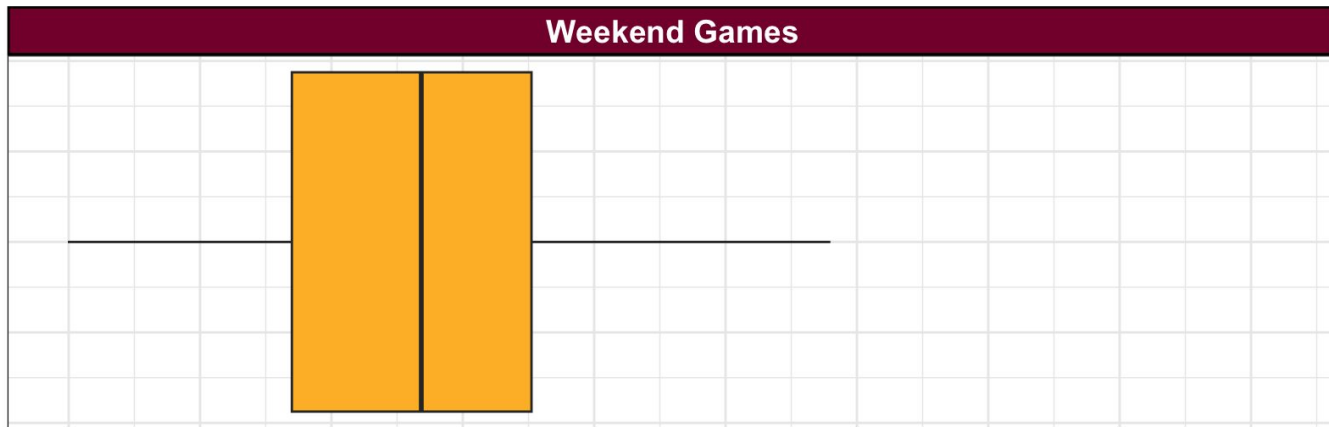
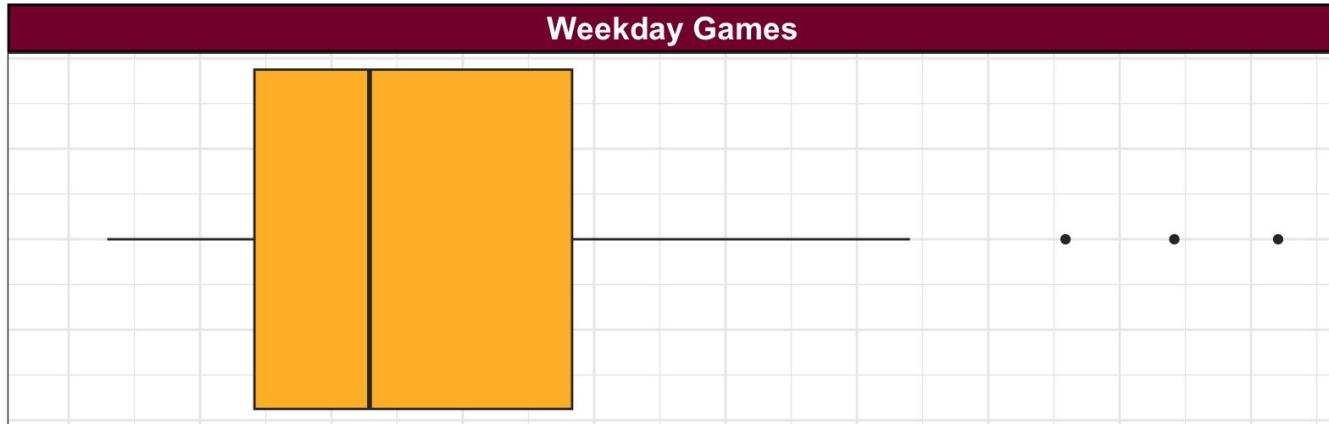
INITIAL EDA

- How do we accurately represent each game's ticket Pacing Line based off historical data?
 - One Approach: Create an expected ticket Pacing Line for each game using time series modeling
 - Alternative Approach: Determine what variable primarily affects ticket sales and build ticket Pacing Lines off of that, utilizing historical averages

Higher Tiered Games Tend to Sell More Tickets



Weekend Games have a Minimal Effect on Ticket Sales



1000 1500 2000 2500 3000 3500 4000 4500 5000 5500

Tickets Sold Each Game

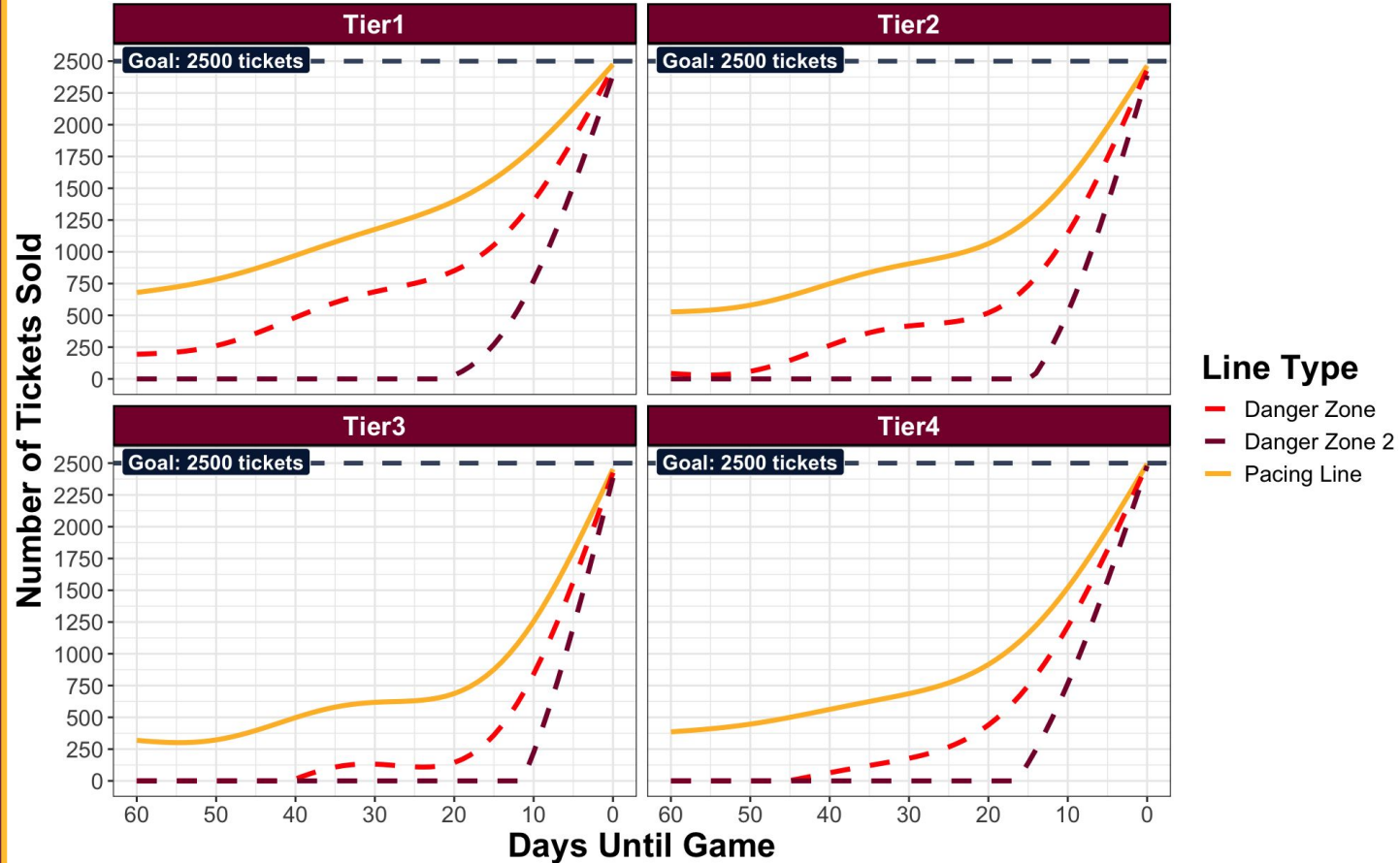
PACING LINE CREATION

- Found ticket sales primarily differ mostly for tiers
 - Weekend, giveaways, etc. had minimal effect on ticket sales
 - Tiers should account for weekend and opponent anyway (*Levicki, 2024, pp. 13-14*)
- Wanted to create a ticket Pacing Line for each tier group
- Our Process:
 - Used cumulative tickets sold for each game
 - i.e. running total of tickets sold as time goes on
 - Found average ticket sales from day 60 to 10 for every 5 days for each tier
 - Manually set day 0 to 2500 tickets
 - Ensures that Cavs will meet their per game goal
 - Fit a Generalized Additive Model (GAM)
 - Flexible line based on average ticket sales and per game goal

DANGER ZONE CREATION

- Our Process (completed for each Tier Group):
 - Shifted each Pacing Line point down by a specific number of tickets from day 60 to 0, dependent on the Danger Zone
 - 500 ticket shift corresponds to **Danger Zone 1**
 - Represents 20% error of the 2500 ticket goal
 - 750 ticket shift corresponds to **Danger Zone 2**
 - Represents 30% error of the 2500 ticket goal
 - Manually set day 0 to 2500 tickets
 - Fit another GAM from this data
 - Any negative predicted ticket values were adjusted to 0
 - Ticket sales cannot be less than 0

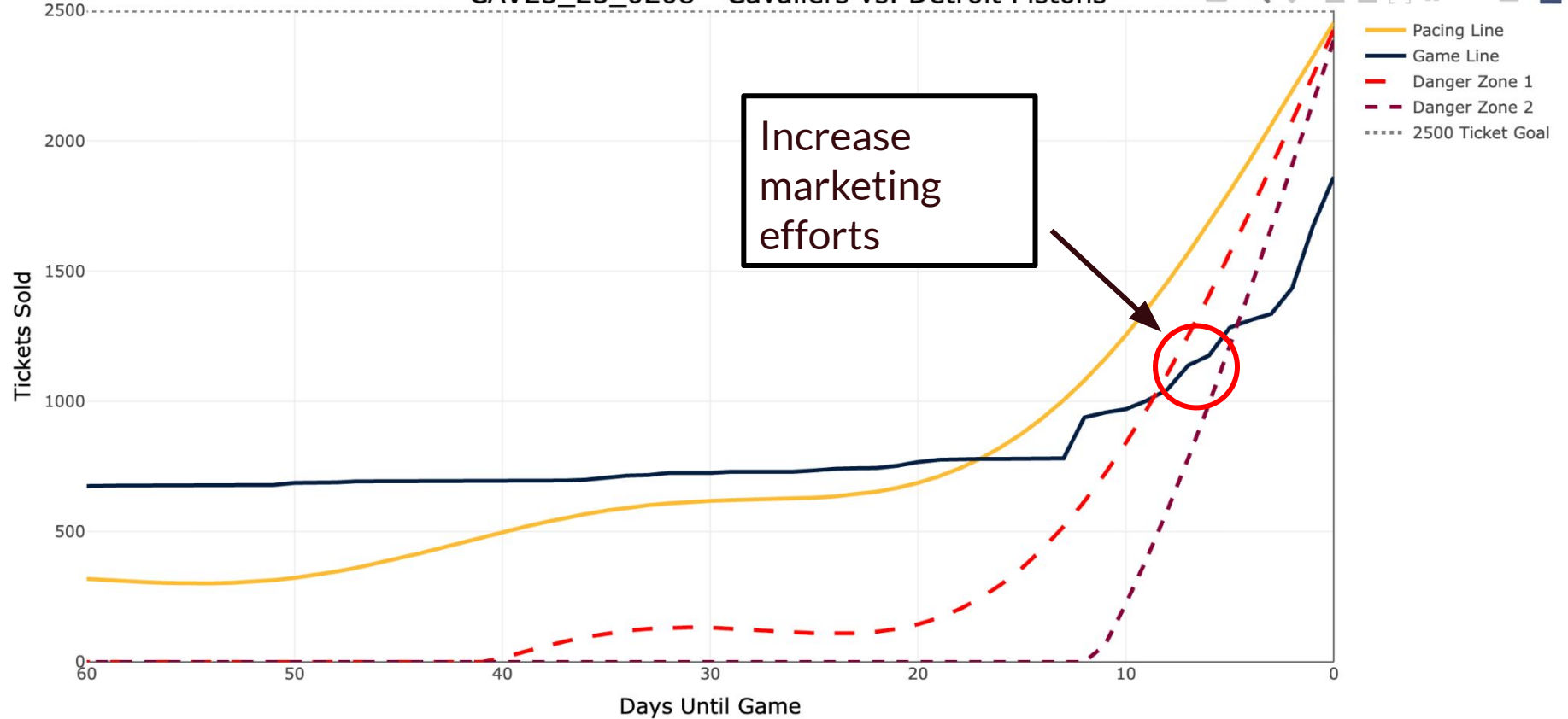
Each Tier has a Different Pacing Line Based on Historical Sales



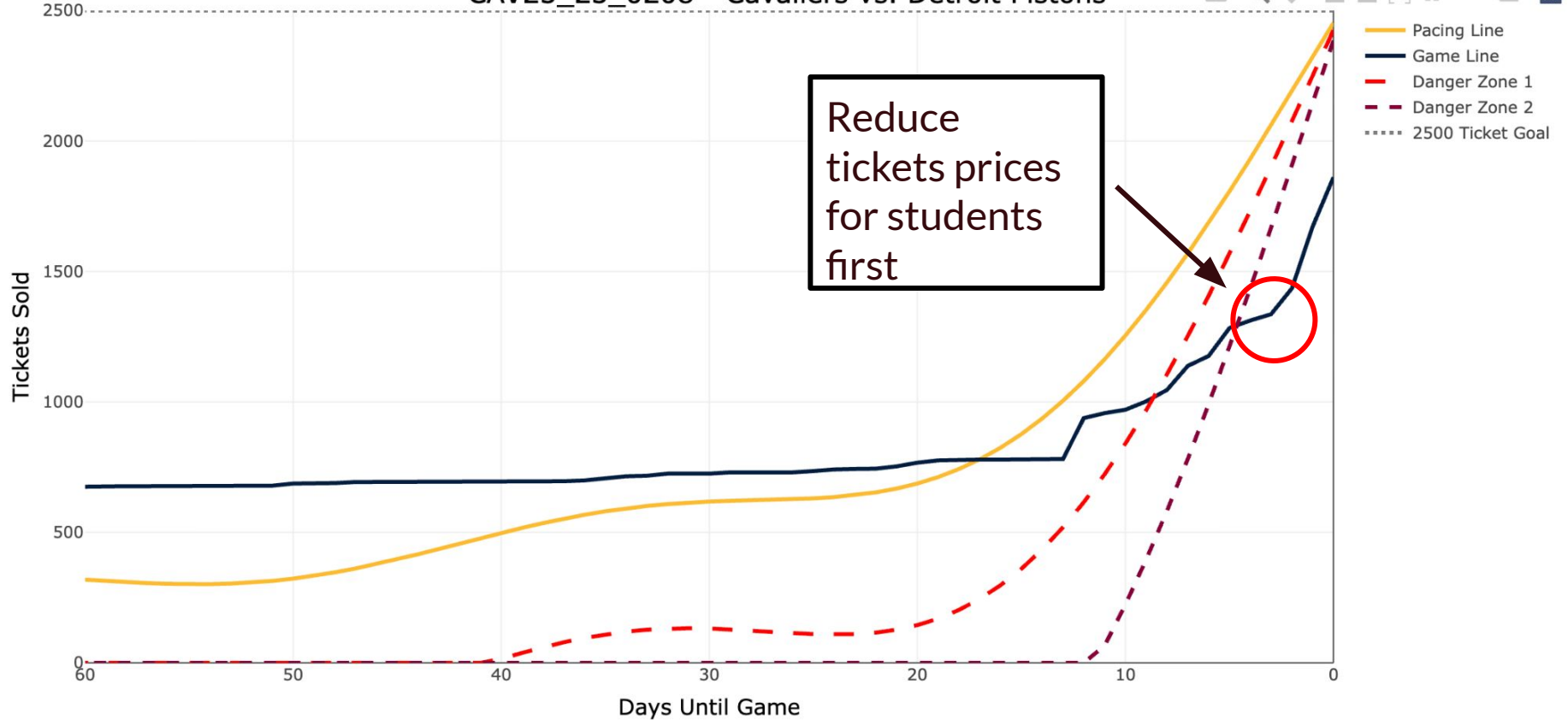
KEY CHARACTERISTICS OF APPROACH

- All line ensure that final ticket sales will be approximately 2500 tickets
- Lines become stricter as the game gets closer
- **When to panic?**
 - When the cumulative ticket sales for a game fall below **Danger Zone 1**, increase marketing through emails and social media avenues
 - When they fall below **Danger Zone 2**, reduce ticket prices, first for students and then groups
 - Due to the nature of the line, **Danger Zone 2** offers a large margin of error before dropping prices is necessary
 - Ensures least amount of revenue is lost

CAV23_23_0208 - Cavaliers vs. Detroit Pistons



CAV23_23_0208 - Cavaliers vs. Detroit Pistons



03

MODELING TICKET SALES

MODELING PROCESS

- **GOAL:** Predict ticket sales for the Cavs game against the Hawks on November 2, 2025
 - Have data from 20 days out until the game
- **Variables Used**
 - Cumulative Tickets Sold on Day 20
 - Very predictive of final day ticket sales
 - Tier Group
- **Variables Not Used**
 - Weekend:
 - Surprisingly was not predictive of ticket sales (high p-value)
 - Giveaway, Theme, and Special Jersey:
 - Were not very predictive and were NA for the Hawks game
 - Opponent:
 - Too few observations

MODELING SELECTION & RESULTS

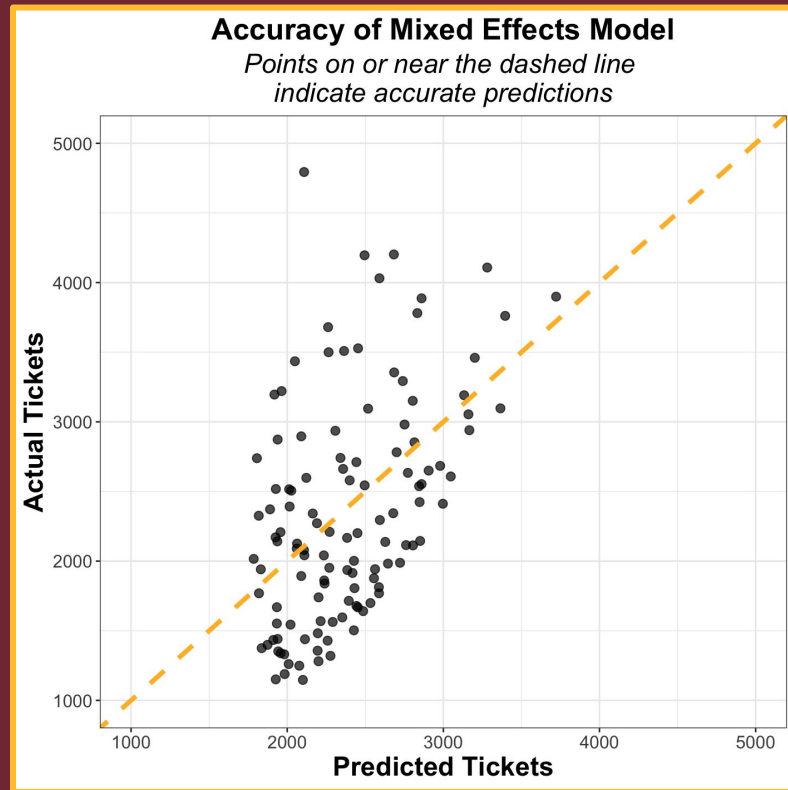
- Utilized 5-fold cross validation to test models
 - XGBoost, GAM, Mixed Effects, and Random Forest
- Evaluated models based off of RMSE

RMSE of Models			
<i>SD of Tickets Sold is 890.02</i>			
XG Boost	GAM	Random Forest	Mixed Effects
840.13	817.8	829.97	795.78

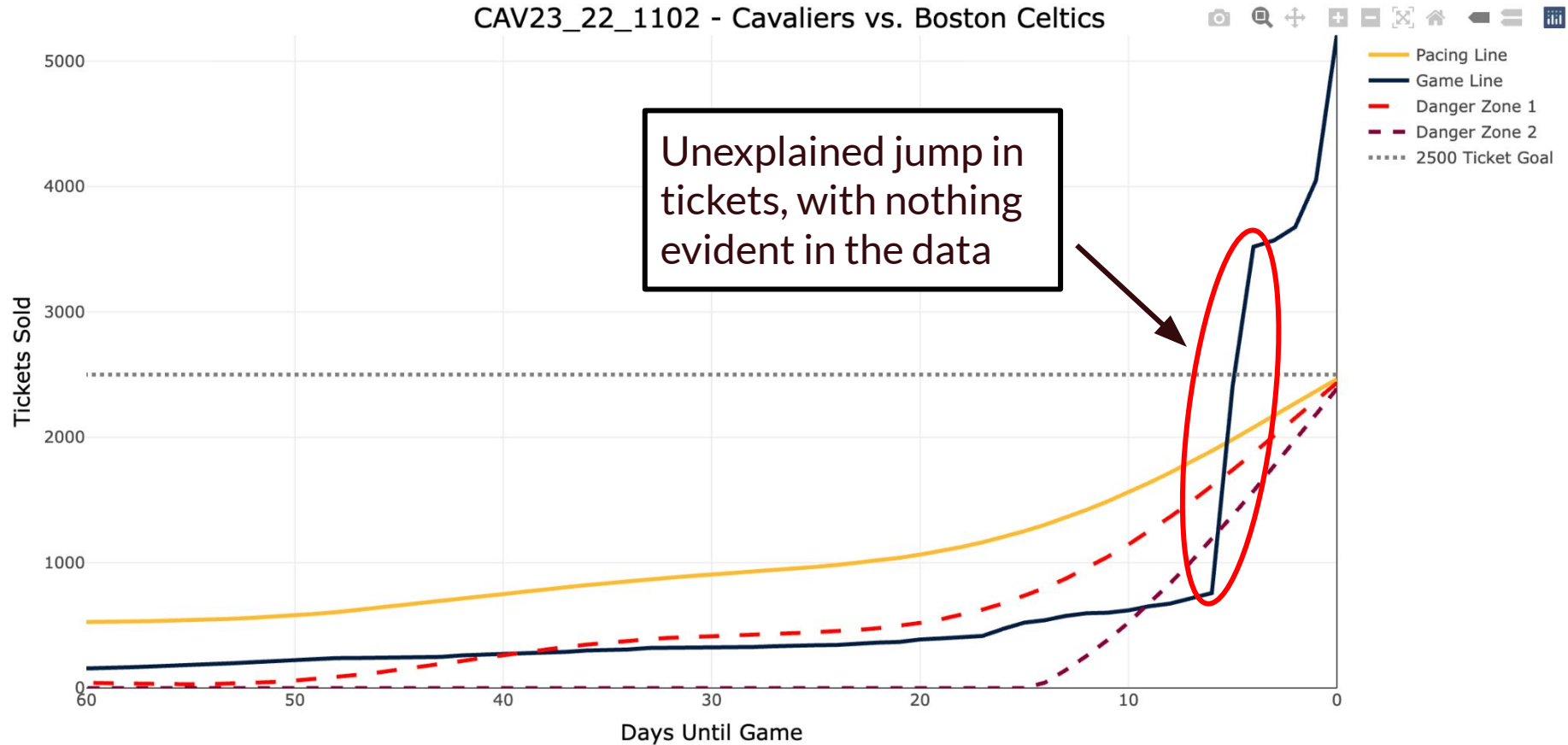
- Mixed Effects Model performs the best
 - Intercepts vary by tier group
 - Tickets sold by day 20 is the primary predictor

ACCURACY OF MODEL

- Mixed Effects Model gives roughly a 10% improvement over random variation
- Still, a RMSE of 795 is a lot of error
- Why?
 - Ticket sales see massive jumps closer to the game (see next slide)
 - Predicting from 20 days out is tough
 - Missing vital predictors
 - Price of tickets, opponent winning percentage, weather, star opponent playing (LeBron, Giannis, Kyrie, etc.)
 - We do not know when the Cavs had marketing campaigns or reduced ticket prices



CAV23_22_1102 - Cavaliers vs. Boston Celtics



SINGLE GAME PREDICTION

- November 2, 2025: Cavs vs. Hawks
 - Final ticket prediction: 2009 tickets
- Be wary of the results
 - Accuracy of model is not high
 - Missing important predictor variables
 - Predicting from 20 days out is tough due to variability of ticket buyers
- Bottom Line:
 - This prediction can be used as a general baseline for staffing/arena needs and is better than randomly guessing the total number of tickets



04

CONCLUSIONS & RECOMMENDATIONS

DASHBOARD

- Created a dashboard to visually summarize results and see ticket sales in real time
- Key Features
 - **Dropdown Menu:** visualize Pacing Lines for each game
 - **Hover Text:** Hover over each point to see key information about the line
 - **Line Description Table:** Lays out what each line represents, so it's easy for business professionals to interpret
- Dashboard Link:
 - <https://lukesnavely.shinyapps.io/Cavs2025/>

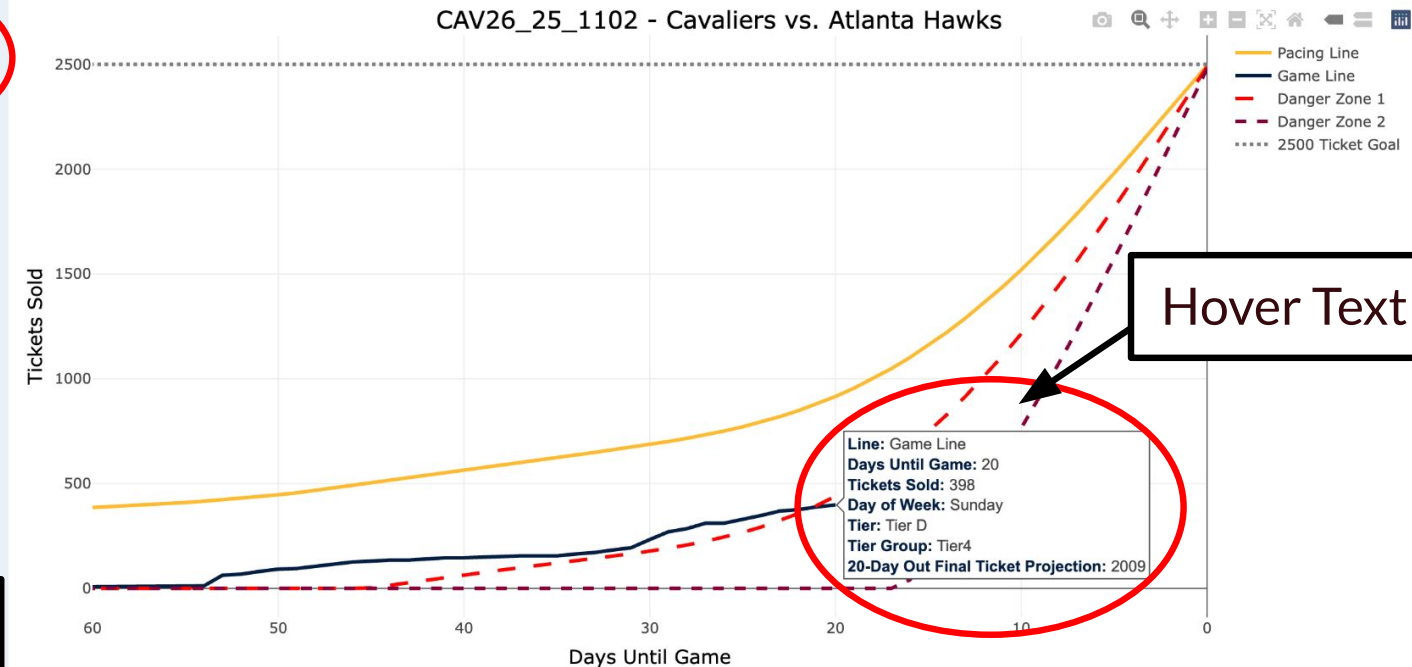
Game Analysis

Select Game:

CAV26_25_1102 - Cavaliers
vs. Atlanta Hawks

Dropdown

Line Description
Table



Hover Text

Descriptions of each Line

PACING LINE

Reaches the per game ticket sales goal of 2500 and mimics real ticket-buying behavior

GAME LINE

Shows actual cumulative ticket sales for each day

DANGER ZONE 1

When the game line is below Danger Zone 1, heavy marketing is suggested

DANGER ZONE 2

When the game line is below Danger Zone 2, discounted tickets are suggested

LIMITATIONS

- By nature, the forecasting model would be more accurate if we predicted closer to the game, such as 7 days out and not 20 days out
 - Ticket sales are **VERY** variable, due to last minute decisions by buyers
 - Primary reason for large RMSE of model
- From historical data, we don't know what marketing campaigns have gone out and when ticket prices were reduced
 - Causes unpredictable spikes in ticket sales without proper context
- Need a better description of what factors are included in the tier ranking
 - Assumed this variable took weekend games/opponents into account, but is this fully true?
- There were several NAs in the data set for giveaway, theme, and special jersey, but NA did not mean an absence of these traits
 - The game we were trying to predict had NAs for these categories

RECOMMENDATIONS

- Danger Zones
 - **Case 1:** The game line is below Danger Zone 1
 - Send out marketing emails or conduct an advertising campaign on social media
 - **Case 2:** The game line is below Danger Zone 2
 - Reduce ticket prices, beginning with students and then groups
- Ticket goals should differ for games and should not be set at 2500 always
 - Example: Lakers will bring in more fans than the Hawks
 - Ticket goals should vary by tier
- For more accurate predictions, model tickets sold closer to the game and include key missing variables (as previously described)
- Plan for about 2000 fans in attendance for Cavs vs. Hawks on November 2, **if and only if** there's no marketing campaigns and no reduction in ticket prices

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